Section 5.2: Finding Sides and Word Problems

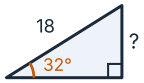
MATH 130 · Friday, November 6 · Quiz 10 on Wednesday, November 11

Name _____

> Pick the ratio that uses the side you **know** and the side you **want**: $\sin\theta = \text{opp/hyp}$, $\cos\theta = \text{adj/hyp}$, $\tan\theta = \text{opp/adj}$.
> Write the equation, then solve. If the unknown ends up in the denominator, divide. Degree mode on, and round to two decimal places at the end.

1 · A missing side

Model (a) The angle is 32° and the hypotenuse is 18. Find the side opposite the angle.
(b) The angle is 54° and the adjacent side is 11. Find the hypotenuse.

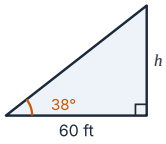


1–2 **1.** The angle is 41° and the hypotenuse is 25. Find the adjacent side.
2. The angle is 27° and the opposite side is 8. Find the adjacent side.

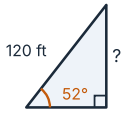
2 · One right triangle in a word problem

Sketch it, label the angle and the known side, mark the side you want, then choose sin, cos or tan.

Model From a point 60 ft from the base of a tree, the angle of elevation to the top is 38°. How tall is the tree?



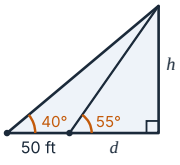
3 A kite string 120 ft long makes a 52° angle with the ground. How high is the kite?



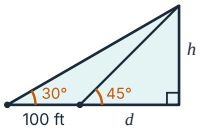
3 · Two right triangles

Write one equation for each triangle using the same height h, then solve them together.

Model From one point the angle of elevation to the top of a tower is 40°. From a point 50 ft closer it is 55°. Find the tower's height h.



4 From one point the angle of elevation to the top of a building is 30°. From a point 100 ft closer it is 45°. Find the building's height.



Common errors: using the hypotenuse with tan; a calculator in radian mode; rounding the first equation before solving the second; leaving off units.